

The Roles of Transmission and Long-Duration Energy Storage in Balancing High-Renewable Power Systems

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1. Abstract

As global energy systems increasingly rely on variable renewable resources in the pursuit of decarbonization, flexible resources are needed to smooth fluctuations in renewable generation. Leading flexible solutions include transmission expansion and energy storage, particularly long-duration energy storage (LDES). While it is generally understood how either solution can help smooth gaps in renewable power generation, their interaction in future high-renewable systems remains underexplored. This study examines how LDES interacts with transmission expansion in future high-renewable energy systems. It explores these interactions using an open-source capacity expansion model (PyPSA) to simulate high-renewable energy system buildout scenarios that include both LDES deployment and transmission expansion, comparing and analyzing scenario outcomes while varying renewable correlation, storage duration, and emissions constraints. Results show that increasing LDES duration reduces system costs and curtailment while shifting the system more toward temporal balancing, changing the balance between spatial and temporal flexibility. However, both transmission and firm capacity remain necessary across all scenarios. These findings emphasize the need for coordinated deployment of storage and transmission to ensure reliability in high-renewable systems.

2. Introduction

2.1 Background and Context

As global energy systems decarbonize, the share of variable renewable energy (VRE) sources in electricity generation is expected to rapidly rise over the coming years; wind and solar alone are forecast to account for 25% of global electricity generation by 2028 (International Energy Agency, 2023a). Higher levels of VRE create growing reliability and flexibility challenges (Jenkins et al., 2018; Sepulveda et al., 2018). The primary challenge is in meeting load demand during lulls in renewable energy generation; in other words, filling in the gaps when the sun is not shining and the wind is not blowing. Notably, this challenge presents itself across timescales, as wind and solar output varies not just hour-to-hour and day-to-day, but also seasonally (Lund et

al., 2005). To solve these challenges, so-called “firm” resources are needed that can provide power at any time to meet shortfalls in variable renewable generation. In many decarbonization pathways studies, it has been shown that firm low-carbon resources are needed to keep energy costs reasonably low and to prevent excessive VRE capacity overbuilds and subsequent curtailment (Sepulveda et al., 2018). However, the use of these low-carbon resources becomes increasingly untenable as carbon budgets become smaller, and other resources have been shown to provide many of the same flexibility benefits while keeping system costs low. These include resources that can shift energy across time or space, reducing the need for dispatchable generation while maintaining reliability in high-renewable systems (Jenkins et al., 2018).

Key flexibility options include transmission expansion and long-duration energy storage (MacDonald et al., 2016; Denholm et al., 2021a). While transmission expansion is a well-established flexibility resource, LDES is relatively new. These technologies are rapidly developing and not yet commonly used in modern electricity systems. Lithium-ion batteries currently dominate energy storage markets, but these systems are typically designed to provide capacity only long enough to satisfy load peaks, around four hours. As energy systems decarbonize, longer durations will become increasingly necessary (Blair et al., 2022). LDES technologies are generally defined as being capable of providing energy for 10 hours or more, though there remain significant variations in this definition, and some technologies may be capable of multi-day or even seasonal storage durations (Denholm et al., 2021b). LDES technologies operate on a wide variety of chemical, electrochemical, mechanical, and thermal energy storage techniques and scientific principles. The most-established LDES technology is pumped-hydro storage; other established LDES technologies include compressed air energy storage (CAES) and molten-salt thermal storage, though these are deployed at a much smaller scale than pumped-hydro (Albertus et al., 2020; U.S. Department of Energy, 2020). Emerging LDES technologies include flow batteries, hydrogen energy storage, liquid air energy storage, iron-air batteries, and gravity energy storage (Denholm et al., 2021b; LDES Council, 2021). LDES technologies are poised for significant growth in the U.S. power sector, with capacities projected to increase five-fold or more between 2020 and 2050 (Blair et al., 2022). At the same time, transmission expansion has been shown to reduce system costs and renewable curtailment in high-VRE energy systems by smoothing variability across geographic regions (MacDonald et

al., 2016; Schlachtberger et al., 2018). This raises important questions about how these two flexible resources interact in long-term energy system planning.

2.2 Problem Statement

As VRE penetration increases, new strategies are needed to maintain electricity system reliability across multiple timescales. High-renewable systems require flexible resources that can balance supply and demand during periods of low renewable generation while avoiding excessive reliance on firm generation resources (Jenkins et al., 2018; Sepulveda et al., 2018). LDES and transmission expansion represent two promising approaches for addressing these challenges. Both resources provide flexibility, but in different ways: storage shifts energy across time, while transmission shifts energy across space. However, the relative deployment of these resources depends strongly on underlying system characteristics, and the conditions under which one becomes more valuable than the other remain insufficiently clarified.

Existing research suggests that both renewable resource correlation across regions and assumptions about storage duration may strongly influence the value of transmission and storage, but these effects are not always cleanly isolated in the literature. Transmission tends to become more valuable when regions can share diverse renewable resources, while longer storage durations may become more valuable when systems must manage prolonged periods of low renewable output. Improving understanding of these relationships is important for utilities, system operators, policymakers, and technology developers making long-term infrastructure decisions in rapidly evolving electricity systems.

2.3 Research Objectives

The objective of this project is to clarify how renewable resource correlation and storage duration affect the relative deployment of long-duration energy storage and transmission expansion in high-VRE electricity systems. While both transmission and storage have been widely studied, less attention has been given to how renewable correlation and storage duration

affect their relative roles in system planning. Establishing a better understanding of these relationships is the central motivation for this project.

To that end, the primary research questions this project seeks to answer are as follows:

- How does renewable resource correlation between regions affect the relative deployment of transmission expansion and long-duration energy storage in a decarbonized high-VRE system?
- How does assumed LDES duration affect the role of storage in balancing high-renewable systems?
- How do these factors influence system outcomes such as total system cost, curtailment, and transmission expansion?

2.4 Scope and Limitations

To investigate these questions within the scope of a one-semester capstone project, this study adopts a simplified modeling framework designed to highlight structural relationships rather than replicate a specific real-world system. The analysis uses a stylized multi-region electricity system in which regions possess distinct renewable resource profiles and are connected by expandable interregional transmission links. This approach allows for the examination of how renewable resource correlation and storage duration influence the deployment of LDES and transmission expansion. The model includes a limited set of technologies representing typical options in high-renewable systems, including wind, solar, short-duration battery storage, LDES, firm backup generation, and transmission expansion. Technology representations are necessarily simplified to keep the model manageable and interpretable within the scope of this project.

While this stylized approach provides a useful experimental environment for examining LDES-transmission interactions, it also contains several limitations. The model does not attempt to replicate the full complexity of real-world transmission networks, energy markets, or system operations. In addition, technology cost assumptions and renewable resource profiles will rely on publicly available datasets and literature values, which may introduce uncertainty. As such, the results should be interpreted as illustrative of structural relationships rather than as forecasts.

Despite these limitations, a simplified modeling framework is appropriate for the primary

objective of this project: examining how renewable correlation and storage duration shape the balance between spatial and temporal flexibility in high-VRE electricity systems.

3. Literature Review

3.1 High-VRE Systems and the Need for Flexibility

Prior literature consistently finds that high shares of variable renewable energy introduce significant reliability and flexibility challenges in power systems. As wind and solar generation increase, maintaining balance between supply and demand becomes more difficult due to their variability and limited dispatchability. Jenkins et al. (2018) identify this need to reliably meet demand during periods of low renewable output as the central challenge of high-VRE systems across a broad range of deep decarbonization studies. They find that curtailment rises nonlinearly as VRE penetration increases and argue that some combination of firm low-carbon resources, long-duration storage, transmission, and other flexible resources is required to keep system costs low. Sepulveda et al. (2018) reach a similar conclusion, finding that systems relying heavily on wind and solar face increasing costs when firm low-carbon resources are excluded due to substantial renewable overbuilds and the difficulty of meeting peak demand during periods of low renewable output. Holttinen et al. (2011) find that large shares of wind power create challenges not only for energy balancing but also for system operation and planning, which may be mitigated through aggregation across larger balancing areas. These studies and others suggest that deep decarbonization and large VRE deployments increase the value of flexible resources that can shift energy across time or space.

3.2 The Emerging Role of Long-Duration Energy Storage

Long-duration energy storage has received increasing attention as a potential means for balancing high-renewable systems over longer timescales than conventional lithium-ion batteries can economically serve. Albertus et al. (2020) provide a useful overview of LDES applications, economics, and technologies, emphasizing that LDES use-cases differ substantially from short-

duration batteries. Whereas short-duration batteries typically provide short-term arbitrage and peak-shaving services, LDES can provide multi-day balancing and reliability support, reducing renewable overbuild and curtailment in high-VRE systems.

Definitions around what storage durations are defined as LDES vary substantially in literature; Denholm et al. (2021b) note that defining LDES is complicated and involves many factors, with durations in the literature ranging from 8 hours to multiple days and even seasons. This discussion highlights that LDES is not a single technology, but a family of technologies with various durations, efficiencies, and cost structures. This point is relevant for this study, which is motivated in part by concerns that many modeling exercises oversimplify LDES.

The economics and design of LDES in decarbonized systems have been studied by many authors. Sepulveda et al. (2021) show that the value of LDES depends strongly on the technical characteristics of the specific technology, particularly cost, efficiency, and duration, and that the optimal LDES design is dependent on system characteristics. Hunter et al. (2021) compare LDES and other flexible generation resources in high-VRE grids and find that LDES can be valuable, but only under specific cost and performance assumptions. Jenkins and Sepulveda (2021) argue for more robust research and innovation around LDES, reflecting the view that longer storage durations will be increasingly important under deep decarbonization.

Despite significant study, there remains much uncertainty about the conditions of LDES value. In a literature review focused on the relationship between VRE penetration and LDES market creation, Selänniemi et al. (2024) find substantial inconsistency in reported “tipping points” at which LDES becomes commercially viable. With VRE penetration levels ranging from 20% to 100%, they conclude that no generalizable threshold exists in the literature. Schleifer et al. (2025) find that the relative value of longer storage durations is region-specific and often peaks at intermediate durations of 20-40 hours. These studies suggest that the value of LDES is highly sensitive to assumptions about system structures.

3.3 Transmission Expansion in High-Renewable Systems

Transmission expansion provides a key form of system flexibility by enabling the transfer of energy across regions, allowing power systems to take advantage of geographic diversity in

renewable resource availability. By aggregating generation across larger areas, transmission reduces variability, curtailment, and reliance on local backup resources. Holttinen et al. (2011) find that high wind penetrations can be more efficiently accommodated with aggregation across larger areas. Likewise, MacDonald et al. (2016) show that large-scale transmission can lower system costs and enable very high renewable penetration by expanding access to geographically diverse resources.

In capacity expansion studies, transmission is consistently found to reduce the need for local backup resources, curtailment, and renewable overbuild. Schlachtberger et al. (2018) utilize a large-scale European model built in PyPSA to find that transmission expansion can lower total system costs in high renewable scenarios, reducing the amount of required backup and storage. In policy-constrained scenarios, they demonstrate that as emissions limits tighten, storage begins to replace gas turbines as a source of flexibility, suggesting that the relative roles of transmission and storage depend not only on technology costs but also on decarbonization constraints. Conlon et al. (2019) similarly show that both storage and new transmission can contribute to meeting renewable generation targets in regional planning, but that their relative value strongly depends on local system conditions. The NREL Storage Futures Study also points to the importance of transmission, though it notes that transmission deferral value is difficult to isolate and highly region-specific (Blair et al., 2022). This finding demonstrates that the specific interaction of transmission with storage is difficult to separate from the other effects that transmission has on system design. This observation undergirds a key motivation for this project.

3.4 Storage–Transmission Interactions

The most relevant literature for this project is the smaller body of work that examines storage and transmission jointly. Neetzow et al. (2018) provide perhaps the clearest conceptual framing, using a stylized theoretical model to show that storage and transmission are not generally either substitutes or complements; rather the nature of the relationship varies by region and system conditions. This result challenges claims that one resource will universally dominate the other.

Among more recent studies, Staadecker et al. (2024) is quite relevant. Using a detailed Western Interconnection model including both LDES and transmission, they find that transmission

constraints can significantly increase the value of storage in some load zones, while unlimited transmission changes the distribution of storage and generation resources more than it changes total storage deployment. They also conclude that storage can reduce the need for transmission expansion, but that the relationship is highly region-specific. Their findings underscore both the policy relevance of transmission barriers and the difficulty of drawing broad conclusions from high-detail regional models alone.

Sepulveda et al. (2018) provide another relevant perspective. In their deep decarbonization scenarios, long-duration storage does not fully substitute for firm low-carbon resources, but transmission expansion does reduce system costs and renewable overbuild when firm resources are unavailable. This result suggests that both transmission and storage can relieve some of the pressures imposed by high renewable penetration, but that the magnitude of their contribution depends strongly on what other resources are available. More broadly, the literature points to a recurring conclusion: storage and transmission interact in meaningful ways, but those interactions remain highly contingent on assumptions about cost, geography, transmission buildability, and storage duration.

3.5 Remaining Gaps and Intent of This Study

The existing literature leaves several points unresolved. While many studies show that transmission and storage both play important roles in high-renewable systems, fewer systematically isolate the conditions under which their relative importance changes. Neetzow et al. (2018) establish conceptually that storage and transmission do not have a fixed relationship, and Staadecker et al. (2024) examine this interaction in a detailed regional model, but there remains limited work that isolates key drivers of this interaction in a controlled setting. Two factors appear repeatedly in the literature but are rarely examined in combination in a simplified framework: renewable correlation across regions and storage duration. Many studies continue to represent LDES using coarse duration bins, often as a 10-hour or 100-hour resource, even though recent work suggests that intermediate durations may materially affect system outcomes (Denholm et al., 2021a; Schleifer et al., 2025). At the same time, while the importance of transmission is well established, its interaction with storage is often difficult to isolate because

transmission value is highly region-specific and confounded by other system characteristics (Blair et al., 2022).

This study builds on these insights by explicitly isolating these two structural variables within a stylized multi-region capacity expansion model. Rather than attempting to replicate a specific real-world system, the analysis is designed to test how spatial variability (through renewable correlation) and temporal flexibility (through storage duration) jointly shape system outcomes. In doing so, the study advances the literature by moving from general observations that storage and transmission interactions are “scenario-dependent” toward a more structured understanding of how and why those dependencies arise. The goal is not to challenge prior findings, but to clarify the mechanisms through which key system characteristics influence the balance between spatial and temporal flexibility in high-renewable power systems.

4. Conceptual Framework

4.1 Spatial and Temporal Flexibility

Transmission and long-duration energy storage provide two distinct forms of flexibility in high-renewable power systems. Transmission enables spatial balancing by moving electricity across regions and allowing systems to take advantage of geographic diversity in renewable resource availability. LDES enables temporal balancing by shifting energy across time, storing excess generation during periods of surplus and discharging during periods of deficit. Both resources can reduce curtailment and improve system reliability, but they operate through different physical mechanisms.

Building on the gaps identified in Section 3.5, this study evaluates how the relative deployment of transmission and LDES changes across scenarios that vary by renewable correlation and storage duration. Rather than treating transmission and LDES as universally substitutive or complementary, the analysis focuses on how systems balance spatial and temporal flexibility under different conditions. In this framework, scenarios in which increased deployment of one resource is associated with reduced deployment of the other can be interpreted as exhibiting substitutive behavior, while scenarios in which both expand together can be interpreted as

complementary. This distinction is used as a lens for interpreting results, rather than as the central organizing concept of the study.

4.2 Spatial Variability and Renewable Correlation

In interconnected power systems, the degree of correlation between renewable resources across regions plays a key role in determining how effective transmission is at reducing variability. Wind and solar generation profiles are not perfectly correlated across space, and this spatial diversity allows transmission to smooth aggregate variability by pooling resources across regions (Holttinen et al., 2011; International Energy Agency, 2023b).

When correlation between regions is low, transmission can reduce variability more effectively by enabling geographic diversification, lowering curtailment and improving resource utilization. When renewable generation is highly correlated across regions, this benefit is reduced, since periods of surplus and deficit tend to occur at the same time across the system. In these cases, temporal balancing becomes more important, increasing reliance on storage to shift energy across time rather than across space (Shaner et al., 2018).

At the same time, spatial aggregation does not eliminate variability entirely, particularly during extended periods of low renewable output across large geographic areas. This implies a continued role for storage even in highly interconnected systems. Renewable correlation therefore shapes how systems balance spatial and temporal flexibility, rather than determining a fixed relationship between transmission and storage.

4.3 Temporal Variability and Storage Duration

Power systems with high shares of variable renewable energy exhibit variability across multiple timescales, requiring different balancing strategies. Short-term variability is driven primarily by diurnal cycles, particularly for solar generation, while longer-term variability arises from multi-day weather patterns and seasonal mismatches between supply and demand (Denholm et al., 2021a).

Short-duration storage technologies, dominated by lithium-ion batteries with durations of around four hours, are well-suited to managing intra-day variability by shifting solar generation from midday to evening peak demand periods (Blair et al., 2022). However, as renewable penetration increases, longer-duration imbalances become more prominent, and short-duration storage alone becomes insufficient to maintain system reliability.

Long-duration energy storage is uniquely capable of addressing multi-day and seasonal imbalances by shifting energy across extended time periods. Prior work shows that as systems approach high levels of renewable penetration, the value of resources capable of sustained energy delivery over long durations increases, particularly during prolonged periods of low renewable output (Sepulveda et al., 2018; Denholm et al., 2021a). While such events may be infrequent, their system impact is significant, making them a key driver of long-duration flexibility needs.

Building upon these findings, this study examines how increasing storage duration affects system outcomes, including whether longer-duration storage displaces short-duration storage and reduces reliance on other balancing resources.

4.4 Expected System Tradeoffs

The primary tradeoffs examined in this study concern the balance between spatial and temporal flexibility in high-renewable systems. Low renewable correlation should increase the usefulness of transmission by allowing regions to share less-synchronized renewable output. Higher renewable correlation should reduce the benefits of spatial smoothing and increase the usefulness of storage that can shift energy across longer periods. Storage duration introduces a second tradeoff. Longer-duration storage increases the amount of energy available per unit of storage power, which may reduce curtailment and lower system costs, but it may also change the system's use of transmission, short-duration batteries, and firm backup capacity. The analysis therefore focuses on how renewable correlation and storage duration affect system cost, curtailment, transmission expansion, storage deployment, and generation mix.

4.5 Hypotheses and Implications for Scenario Analysis

The central hypothesis of this study is as follows:

In high-renewable power systems, the balance between transmission and long-duration energy storage is shaped by the relative importance of spatial and temporal variability, such that systems shift toward spatial flexibility under low renewable correlation and toward temporal flexibility under high correlation and longer storage durations.

This overarching hypothesis is supported by the following sub-hypotheses:

- Sub-hypothesis 1: Renewable correlation effect

As inter-regional renewable correlation decreases, transmission becomes more effective at reducing variability, while higher correlation increases reliance on temporal balancing through storage.

- Sub-hypothesis 2: Storage duration and temporal flexibility

Increasing storage duration allows the system to manage longer-term imbalances, leading to a shift away from short-duration storage toward longer-duration solutions.

- Sub-hypothesis 3: Persistence of spatial and temporal flexibility resources

Even with high deployment of either transmission or long-duration storage, systems continue to require a mix of spatial and temporal flexibility.

The spatial-temporal tradeoffs motivate a scenario design that isolates renewable correlation and storage duration as the primary variables of interest. By varying these parameters within a stylized multi-region capacity expansion model, the analysis examines how power systems adjust the balance between spatial and temporal flexibility under different conditions. The analysis places particular emphasis on how changes in storage duration affect system configuration, while renewable correlation provides the spatial context within which these changes occur.

5. Methodology

5.1 Modeling Framework

This project utilizes a modeling framework built around capacity expansion modeling (CEM) and scenario analysis. CEM tools are used in long-term planning for power systems to identify the least-cost mix of resources under various technological and cost assumptions. Inputs to the models include load profiles, generation profiles, cost assumptions, and operational constraints. The models co-optimize investment and dispatch decisions in the context of system operations to minimize overall system cost while meeting demand. CEMs are highly suitable for the analysis of high-renewable scenarios in which there is significant temporal variation in generation, allowing long-term planning outcomes to reflect operational realities and system constraints. Ongoing improvements in capacity expansion modeling have increased temporal resolution and computational tractability, enabling more realistic representation of these dynamics (Chernyakhovskiy et al., 2021).

This project uses the PyPSA capacity expansion model, an open-source python package for power systems analysis developed for planners and researchers (PyPSA Developers, n.d). PyPSA is commonly used in academic literature and is thus an excellent fit for this project. The PyPSA CEM optimizes capacity and dispatch decisions over some time span for a user-defined network of system components, including generators, storage, and transmission links. The objective of the optimization is to minimize total system cost, comprised of capital costs and operational costs, while meeting load demand over the modeled duration. The optimization is subject to numerous constraints, the most important of which are energy balance and capacity constraints but also may include transmission constraints and carbon emissions constraints.

5.2 Model Structure

5.2.1 Regions

The modeled network for this project is composed of three distinct “regions”, which are modeled as *Buses* in PyPSA. Each region has its own distinct load profile and renewable resource profiles,

and each of the three regions is connected to the other two via transmission, or *Links*. The goal of the optimization is to determine the optimal mix of system components in each region, along with transmission capacity between them, to meet the load in each region while minimizing cost.

The regions modeled are “stylized,” meaning they are not meant to represent any real-world region with load and renewable resource profiles unique to that location. Instead, load and resource profiles were chosen for each region with the intent of emulating specific desired regional characteristics while not adhering to strict reality. This and similar approaches are commonly used in literature to examine structural behaviors of power systems that are generalizable to many systems in the real world.

5.2.2 Generators

Each region includes wind, solar, and firm gas *Generators*. Generators in PyPSA are defined by capacity and cost attributes. Each of the generators are assigned an annualized capital cost, and the firm generator is assigned a marginal cost. Marginal costs for wind and solar are assumed to be zero. Wind and solar generators are assigned a time series of capacity factors, while the firm generator is assumed to be perfectly dispatchable. Each generator is set to have expandable capacity, which tells the model solver that the capacity of each generator is a decision variable to be solved for.

5.2.3 Storage

In addition to generators, each region is assigned a pair of *Storage* components, representing short- (battery) and long-duration storage. The key attributes for storage components are cost, capacity, charge/discharge efficiencies, and duration. Annualized capital costs are assigned to each while marginal costs are assumed to be zero. Charge and discharge efficiencies are assumed to be 92% for short-duration storage and 80% for long-duration storage, consistent with typical ranges reported in the literature (Hunter et al., 2021). Short-duration storage is given a duration of 4 hours, while the duration of the long-duration storage is varied across scenarios, with values of 10, 24, 72, and 100 hours, representing a range of potential storage durations for various

LDES technologies. Like the generators, capacity is set to expandable, and the model solves for the optimal capacity of each storage type.

5.2.4 Transmission Links

Transmission is modeled as expandable *Links* between regions, with cost, capacity, and efficiency attributes. For this project, annualized capital costs are derived from literature estimates and marginal costs are assumed to be zero. Transmission efficiency is assumed to be a fixed 97%, reflecting simplified transmission losses in a stylized manner. Transmission capacity is set to expandable and is determined by the model solver. Importantly, these transmission links represent a stylized model, and the PyPSA CEM does not perform detailed power flow analysis when solving for transmission capacities. In addition, real-world distances between regions are not modeled. This is a simplified representation and does not capture distance-specific transmission costs or losses. However, this abstraction is consistent with the study's focus on structural relationships rather than system-specific geography.

5.2.5 CO₂ Emissions Cap

To ensure that the model optimizes for a high-VRE system, a CO₂ emissions cap constraint is applied to the network. Without this constraint, the model optimization would likely rely heavily on firm gas generation to meet reliability needs, obscuring the interactions between LDES and transmission expansion that are central to this study. Each of the system components are assigned an emissions value; for all components except the firm gas generator, this value is assumed to be zero. By limiting total emissions, and thus gas generation, the model is forced to balance variable renewable generation with storage and transmission options, allowing the structural tradeoffs between these resources to emerge clearly.

The emissions cap used is defined relative to a baseline level in which all electricity demand is met by the firm gas generator. The gas generator is assumed to have an emissions intensity of 0.4 tCO₂/MWh, consistent with values reported by the U.S. Energy Information Administration (2021). Baseline emissions are calculated as the product of total annual demand for the modeled

network and this emissions intensity. The emissions cap is then set as a fixed fraction of this baseline. A value of 5% was used for the primary scenario analysis, corresponding to a 95% reduction in emissions from the baseline. This level was chosen to represent a stringent but non-zero emissions constraint, ensuring the system is highly reliant on low-carbon variable renewable resources while still permitting the use of some firm generation for reliability.

5.2.6 Temporal Resolution

The model is optimized over one year of hourly data, yielding 8,760 snapshots. This temporal resolution captures diurnal and seasonal variation that is important for studying storage and transmission behavior under renewable generation variability. All load and renewable profiles are aligned to a common hourly UTC index for the year 2023. Any time-series inputs, including load and renewable capacity factor profiles, are therefore represented at the same hourly resolution.

5.3 Data Inputs

5.3.1 Renewable Resource Profiles

Renewable resource profiles for wind and solar generators are represented as capacity factor time series, with 8,760 values between 0 and 1, representing the capacity factor at each snapshot in time. Wind and solar output profiles for this project were generated using Renewables.ninja, a webtool that allows users to generate simulated hourly power output data from wind and solar power plants located anywhere in the world (Pfenninger & Staffell, n.d.). For this project, outputs were generated for various locations around the Western United States, including coastal California, the desert Southwest, the Pacific Northwest, and the Mountain West. These locations were chosen to generate a selection of realistic resource profiles with differing characteristics, enabling the identification of both highly correlated and highly diverse renewable generation profiles. Data from 2023 was used consistently across all locations to preserve temporal comparability across regions. The application of these profiles to model scenarios is discussed in Section 5.4.

5.3.2 Load

Load data for the model is derived from historical hourly demand data from the California Independent System Operator (CAISO) for the year 2023, providing a representation of real-world electricity demand, capturing diurnal and seasonal variation (California Independent System Operator, 2024). This data is scaled by varying degrees and applied to each of the three regions in the stylized system. Each region's demand thus shares the same temporal structure, but with differing absolute demand levels, representing differences in regional size. This approach preserves realistic temporal dynamics within systems while maintaining simplicity. Using a consistent temporal load shape across regions ensures that outcomes are derived primarily from differences in renewable resource profiles and system configurations, making the results more clearly analyzable and attributable to these differences.

5.3.3 Technology Costs

5.3.3.1 Cost Structure

All technology capital costs used in the model are expressed in terms of overnight capital cost (OCC) and annualized using a fixed discount rate. A discount rate of 7% was used, consistent with financial assumptions commonly applied in energy system modeling studies and in the NREL Annual Technology Baseline (NREL, 2024). In addition, fixed and variable O&M costs are modeled for each technology, and marginal (fuel) costs are modeled for firm gas generation. Marginal costs for wind, solar, and storage are assumed to be zero.

5.3.3.2 Treatment of Uncertainty

Significant uncertainty exists in the future costs and performance characteristics of key technologies, particularly long-duration energy storage and transmission infrastructure. Cost estimates for these technologies vary substantially across studies due to differences in technology maturity, system configuration, geography, and assumptions regarding permitting and deployment constraints. To maintain a manageable and interpretable scenario structure, this

study does not explicitly vary technology costs across scenarios. Instead, representative baseline values informed by recent literature and publicly available datasets are used consistently across scenarios. This approach allows the analysis to isolate the effects of renewable resource correlation and storage duration on system outcomes without introducing additional scenario dimensions related to technology cost uncertainty.

5.3.3.3 Cost Assumptions

Technology cost assumptions are primarily based on recent estimates from the National Renewable Energy Laboratory Annual Technology Baseline (ATB) (NREL, 2024). Utility-scale solar and wind costs are taken from ATB moderate-case estimates, while battery storage costs are derived from ATB projections for 4-hour lithium-ion systems. Natural gas generation is represented as a generic firm backup resource using combustion turbine cost estimates from the U.S. Energy Information Administration Annual Energy Outlook technology cost assumptions (U.S. Energy Information Administration, 2023). This reflects the role of gas generation as a flexible, infrequently used resource rather than a baseload generator. A single representative value is selected for each technology to provide a consistent baseline. Long-duration energy storage costs are more uncertain and are represented using a single baseline value informed by the Long Duration Energy Storage Council (LDES Council, 2021). Transmission costs are also represented using a stylized baseline value informed by International Energy Agency analysis of grid expansion and transmission infrastructure (International Energy Agency, 2023b). Because transmission costs vary widely depending on geography, system configuration, and permitting constraints, this value should be interpreted as a simplified modeling assumption rather than as a site-specific estimate.

Technology	CapEx (\$/kW)	Fixed O&M (\$/kW-y)	Marginal Cost (\$/MWh)	Lifetime (years)	Source
Wind	1,600	32	0	30	NREL, 2024
Solar	1,480	24	0	30	NREL, 2024

Battery (4h)	1,300	30	0	15	NREL, 2024
LDES	1,800	36	0	20	LDES Council, 2021
Transmission	400	8	0	40	IEA, 2023b
Backup (gas)	560	15	45	30	EIA, 2023

Table 1: Technology cost assumptions

5.3.4 CO₂ Emissions

An emissions intensity of 0.4 tCO₂/MWh is assumed for natural gas generation, based on values reported by the U.S. Energy Information Administration for CO₂ emissions from gas-fired electricity generation (EIA, 2021). For all other generators and model components, CO₂ emissions are assumed to be zero. Lifecycle emissions are not considered in this analysis.

5.4 Scenario Design

5.4.1 Variables

This study focuses on two primary scenario variables: renewable resource correlation and LDES duration. These variables are chosen because they directly represent the two core flexibility dimensions explored in the conceptual framework: spatial diversity in renewable generation and the temporal duration of available storage technologies.

5.4.1.1 Renewable Resource Correlation

A primary variable in this study is the degree of correlation in renewable resource availability between regions. To study the impact of this correlation, two cases are defined for the scenario analysis: a high-correlation case and a low-correlation case. To construct these cases, wind and solar resource profiles were first generated for a set of candidate geographic locations across the Western United States using Renewables.ninja. Each candidate location provides a pair of hourly

wind and solar capacity factor time series. Each modeled region is represented by one of the geographic coordinates, from which both wind and solar profiles are derived. All possible three-region combinations were constructed from these locations. For each region in each combination, a composite renewable generation profile was calculated as a sum of the normalized wind and solar profiles. Correlations were then calculated between the composite renewable profiles of each region. The average of these correlations was then used as a metric to determine the final two scenario cases. The trio of regions with the highest average correlation is the high-correlation case, while the trio of regions with the lowest average correlation is the low-correlation case. This method maximizes the correlation difference between scenarios as an experimental variable while ensuring data remains realistic. The following table summarizes the actual locations used for each scenario.

Correlation Case	Region 1	Region 2	Region 3	Correlation Score
Low	45.7, -120.7 (Columbia River Gorge, OR)	33.9, -116.8 (San Geronio Pass, CA)	(39.5, -104.5 Colorado Front Range, CO)	0.428
High	32.8, -116.4 (East San Diego County, CA)	35.0, -117.3 (Mojave Desert, CA)	35.5, -118.3 (Tehachapi Wind Area, CA)	0.843

Table 2: Renewable profile locations used for high and low correlation cases

5.4.1.2 LDES Duration

LDES duration is the other primary scenario variable and is modeled with varying discharge durations to represent different classes of storage technologies. Four durations were used: 10 hours, 24 hours, 72 hours, and 100 hours. These values span a range from extended-duration battery systems to emerging and future technologies capable of multi-day storage. Varying storage duration allows the analysis to capture how the role of storage changes as its ability to address long-duration deficits increases.

5.4.2 Scenario Matrix

Every possible combination of the two primary variables was solved independently, resulting in a matrix of 8 primary scenarios. The following table summarizes these.

Scenario ID	Renewable Correlation	LDES Duration
1	Low	10 h
2		24 h
3		72 h
4		100 h
5	High	10 h
6		24 h
7		72 h
8		100 h

Table 3: Primary scenario matrix

In addition to the primary scenario matrix, a limited sensitivity analysis was conducted on the 100-hour LDES scenarios to evaluate the influence of the CO₂ emissions constraint on system outcomes. For both renewable correlation cases, additional scenarios were solved with emissions caps equal to 10% and 20% of the gas-only baseline. This sensitivity analysis was intended to examine how relaxing decarbonization constraints affects the relative deployment of transmission, long-duration storage, and firm generation.

Scenario ID	LDES Duration	Renewable Correlation	CO ₂ Emissions Cap
4*	100 h	Low	5%
9			10%
10			20%
8*		High	5%
11			10%
12			20%

Table 4: CO₂ emissions sensitivity analysis scenario matrix

*The 5% cases are included in the primary scenario matrix and are repeated here as the baseline comparison for the sensitivity analysis.

5.5 Solution and Analysis

5.5.1 Model Outputs

The model produces outputs describing both system investment decisions and operational behaviors. The primary output representing investment decisions is installed capacities for each generating technology (wind, solar, firm gas) in MW, storage capacities in MW and MWh, and transmission capacity between regions in MW. System operation output metrics include total system cost, renewable curtailment, dispatch patterns, storage utilization, and transmission utilization.

5.5.2 Analysis Approach

Results are analyzed through comparison across scenarios to evaluate how system outcomes change as renewable correlation and assumed storage duration are varied. The analysis focuses on the relative deployment of transmission and LDES, as well as associated changes in total system cost, curtailment, and storage utilization. Particular attention is given to how spatial and temporal flexibility requirements shift across scenarios, and to the conditions under which

changes in one resource are associated with increased or decreased deployment of the other. Although the study does not treat substitution and complementarity as its primary framing device, those concepts are still used as an interpretive lens for reading results. Scenarios in which greater deployment of one resource is associated with reduced deployment of the other are interpreted as exhibiting substitutive behavior, while scenarios in which both resources expand together are interpreted as exhibiting complementary behavior. This allows the analysis to remain grounded in the existing literature while keeping the primary emphasis on measurable deployment outcomes and system behavior.

5.6 Limitations

This study has several important limitations that warrant acknowledgement and discussion. First, the modeled system is highly stylized and does not represent any specific real-world power system. While this is intentional and allows for clearer identification of structural relationships, results may not directly translate to specific real-world systems. Second, transmission is modeled using simplified links without detailed power flow analysis, and thus network physics and congestion issues are not accounted for. Third, storage is modeled using fixed-duration representations, which simplifies the behavior of real-world storage technologies that may have more complex behaviors. Finally, cost assumptions are highly uncertain. While representative baseline costs are used to manage this uncertainty, results likely remain sensitive to these assumptions.

6. Results

6.1 Scenario Overview

This section presents results from the 8 primary scenarios, consisting of two renewable correlation cases (“low” and “high”) across four LDES durations: 10, 24, 72, and 100 hours. All scenarios were solved under the same load conditions and a consistent CO₂ emissions constraint corresponding to a 95% reduction from a gas-only baseline. The model determined system

configurations through least-cost optimization, with results reflecting optimal investments in generation, storage, and transmission capacity.

A limited sensitivity analysis extension was also included for the 100-hour LDES case to examine how relaxing the CO₂ emissions constraint affects the relative deployment of LDES, transmission, and backup generation.

Complete results for the study can be found in the Appendix.

6.2 System Cost Outcomes

The results demonstrate a clear pattern with respect to total system cost across the 8 scenarios. System cost declines with increasing LDES duration across both correlation cases, with the largest reductions occurring between 10 and 72 hours. Expressed as percentage reductions from the 10-hour case, system cost declines by approximately 16% in the low-correlation case and 22% in the high-correlation case when LDES duration increases from 10 to 100 hours. This indicates that longer-duration storage provides larger relative cost benefits in the high-correlation system.

Correlation case	10h cost	100h cost	Reduction
Low	\$127/MWh	\$107/MWh	~16%
High	\$155/MWh	\$120/MWh	~22%

Table 5: System cost results summary

The marginal benefit of increasing storage duration appears to diminish at longer durations. While extending duration from 10 to 24 hours yields substantial cost reductions, the additional benefit from 72 to 100 hours is relatively small in comparison. This pattern is consistent across both correlation cases.

At all durations, costs remain more expensive in the high-correlation system than the low-correlation system. However, the cost gap narrows as storage duration increases. In other words, the gains in system cost reduction are larger for the high-correlation system. This suggests that longer-duration storage partially mitigates the additional balancing challenges associated with highly correlated renewable generation.

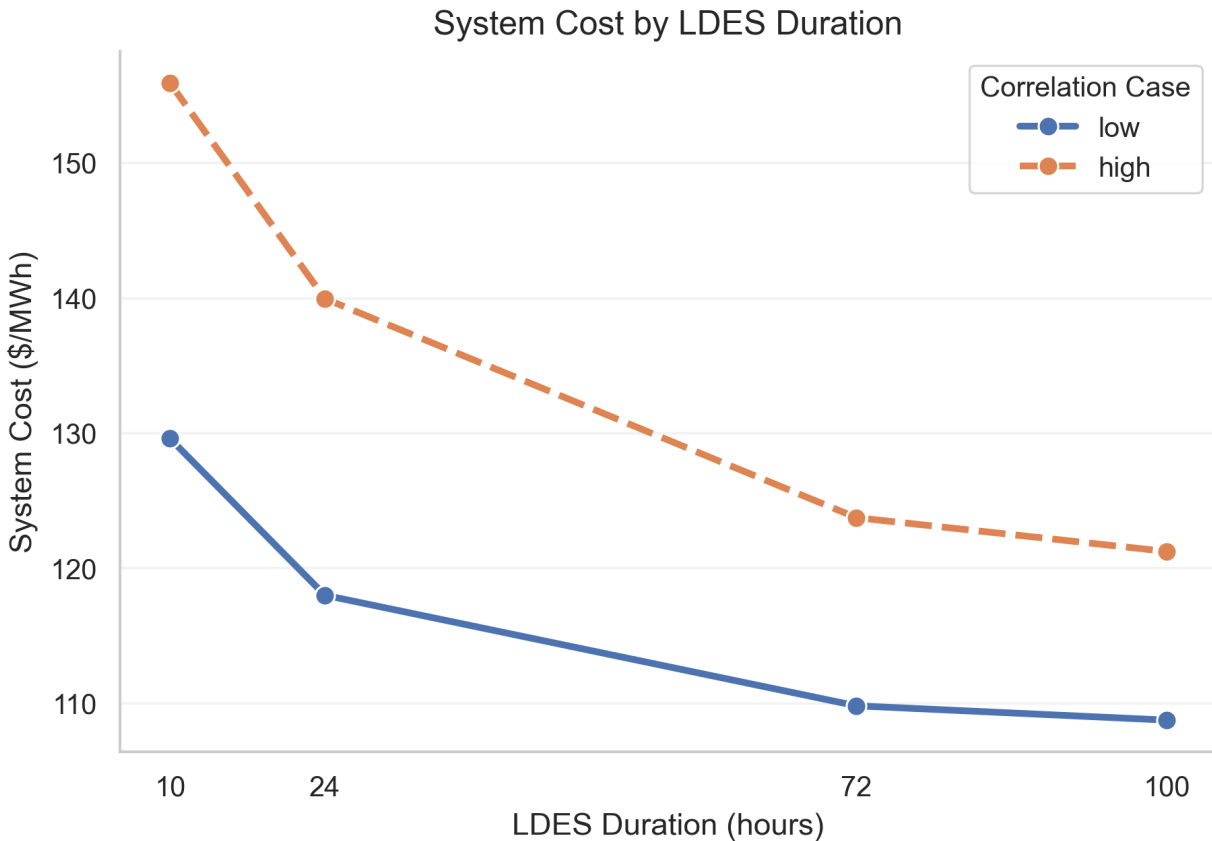


Figure 1: System cost as a function of LDES duration for both correlation cases.

6.3 Curtailment and System Efficiency

Renewable curtailment follows a similar pattern to system cost. In both correlation cases, renewable curtailment decreases as storage duration increases, though the magnitude of this effect differs sharply between the two cases. In the low-correlation case, longer-duration storage

reduces curtailment by roughly 63%, from 109 TWh to 40 TWh. In contrast, the high-correlation system exhibits a more modest reduction, with curtailment decreasing by only about 18%, from 117 TWh to 96 TWh over the same range. Curtailment increases slightly from 10 to 24 hours, and appears flat from 72 to 100 hours, with all the decrease in curtailment occurring when moving from 24 to 72 hours.

Correlation case	10h curtailment	100h curtailment	Reduction
Low	109 TWh	40 TWh	~63%
High	117 TWh	96 TWh	~18%

Table 6: Renewable curtailment reduction from 10h to 100h LDES duration.

These results indicate that longer-duration storage generally improves renewable utilization in both cases but is more effective when renewable generation is less spatially correlated. In the high-correlation case, excess renewable generation occurs more synchronously across regions, limiting the ability of storage alone to absorb excess output. This also suggests that other flexible uses of surplus energy, such as flexible hydrogen production or other controllable loads, could be important complements to LDES in future work.

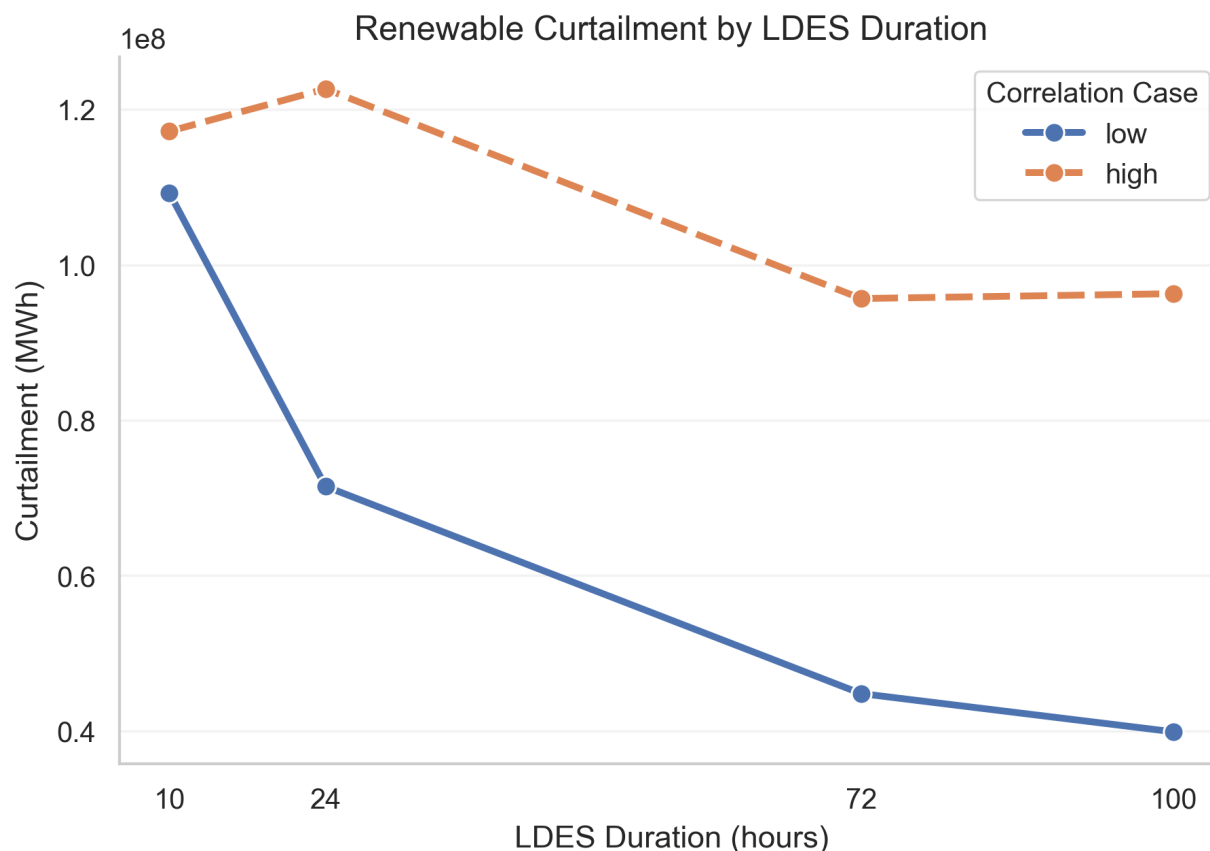


Figure 2: Curtailment as a function of LDES duration for both correlation cases.

6.4 Transmission and Storage Deployment

Transmission and storage deployment show different patterns across the two correlation cases. In the low-correlation system, transmission capacity declines modestly as LDES duration increases, falling from approximately 36 GW in the 10-hour case to 32 GW in the 100-hour case. This suggests some substitution between longer-duration storage and transmission, but the effect is limited. Transmission remains high across all low-correlation scenarios, indicating that spatial diversity continues to provide substantial system value even when longer-duration storage is available.

The high-correlation case behaves differently. Transmission capacity increases from approximately 21 GW in the 10-hour case to 33 GW in the 72-hour case, before falling slightly

to 27 GW in the 100-hour case. This is one of the more important results of the study because it suggests that LDES and transmission can become complementary under some conditions. Rather than simply replacing transmission, longer-duration storage appears to increase the usefulness of transmission by allowing the system to move surplus generation across regions and store it over longer periods.

LDES power capacity declines as duration increases in both cases, while total LDES energy capacity increases. This reflects the structure of fixed-duration storage: longer-duration resources require less discharge capacity to provide the same or greater amount of stored energy. For example, the model builds fewer MW of LDES at longer durations, but because each MW is paired with more hours of storage, total MWh increases substantially. This distinction between storage power and storage energy is important for interpreting the results. The system is not necessarily building more storage capacity in power terms, but it is building much more stored energy.

Overall, the results do not show a simple substitution relationship between transmission and LDES. In low-correlation systems, longer-duration storage modestly reduces transmission needs. In high-correlation systems, longer-duration storage can increase transmission deployment, suggesting complementarity between spatial transfer and multi-day energy shifting.

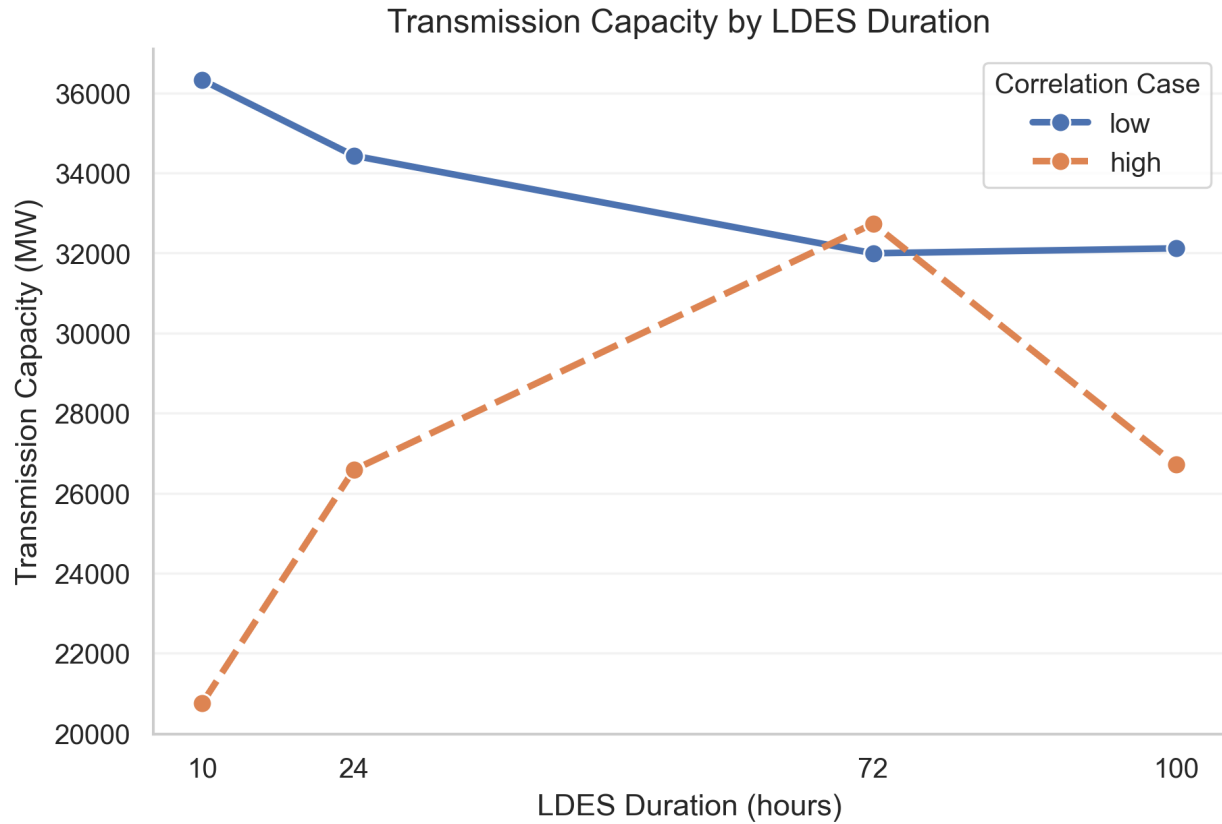


Figure 3: Transmission capacity as a function of LDES duration for both correlation cases.

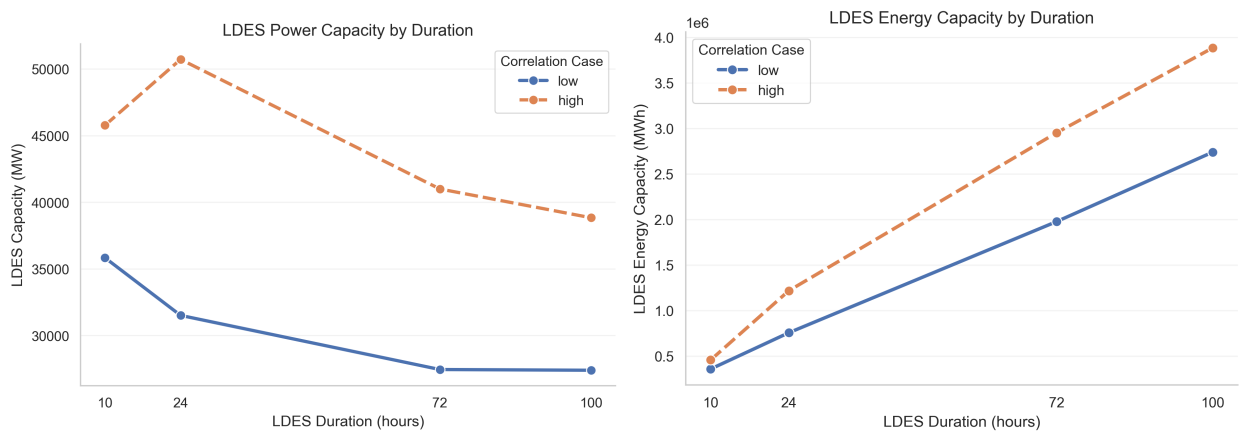


Figure 4: LDES power capacity (left) and energy (right) as a function of LDES duration for both correlation cases.

6.5 Storage Technology Substitution

The results indicate a clear shift in the composition of storage resources as LDES duration increases. In the high-correlation system, short-duration battery storage is deployed at approximately 33 GW in the 10-hour case but is nearly eliminated in scenarios with longer LDES durations. In the low-correlation system, battery deployment is minimal across all scenarios. This pattern suggests that increasing LDES duration allows a single resource to provide both intra-day and multi-day flexibility, reducing the need for dedicated short-duration storage. As a result, the system transitions from diurnal balancing toward longer-duration temporal shifting as storage duration increases.

This result suggests that the main substitution effect in the model is not between LDES and transmission, but between LDES and short-duration batteries. As LDES duration increases, the system appears to rely less on dedicated diurnal storage and more on storage resources capable of covering both daily and multi-day imbalances. This does not necessarily imply that short-duration batteries would disappear in real systems, since the model does not fully represent ramping speed, degradation, ancillary services, or sub-hourly operating needs. However, within the simplified capacity expansion framework, longer-duration storage captures much of the value otherwise provided by 4-hour batteries.

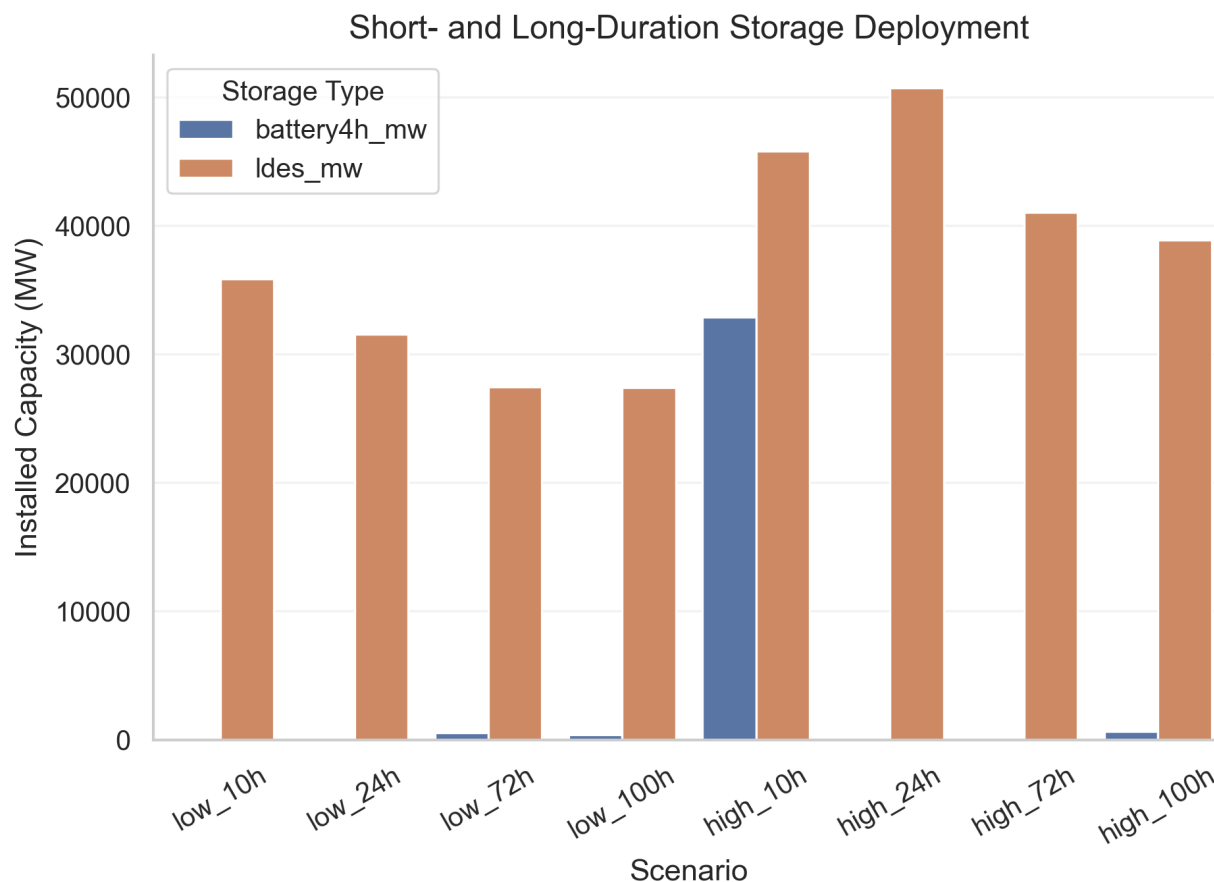


Figure 5: Installed capacity of short- and long-duration storage by scenario.

6.6 Generation Mix and Firm Capacity

The optimal generation mix shifts modestly with increasing storage duration, with the greatest shift occurring between wind and solar generation. In general, longer-duration storage is associated with reduced reliance on solar generation, and in the high-correlation case, increasing reliance on wind generation. This reflects the ability of longer-duration storage to accommodate more variable and less temporally concentrated generation profiles. In the low-correlation case, the relationship is more muted, with decreasing solar generation but stable wind generation as storage duration increases, indicating a system that relies more on spatial balancing. This result may be important for real-world planning because storage duration can affect not only the amount of storage built, but also the preferred renewable generation mix. Longer-duration storage appears to make wind generation more attractive in the high-correlation case by

improving the system’s ability to manage less diurnally predictable output. By contrast, shorter-duration storage pairs more naturally with solar generation because solar surplus is concentrated in daily cycles.

Across all scenarios, a significant amount of backup generation capacity is retained, ranging from approximately 15 to 24 GW. In all scenarios, backup generation energy production is at the maximum allowed by emissions constraints. This indicates that even with substantial deployment of renewable generation, storage, and transmission, firm capacity remains necessary to ensure system reliability.

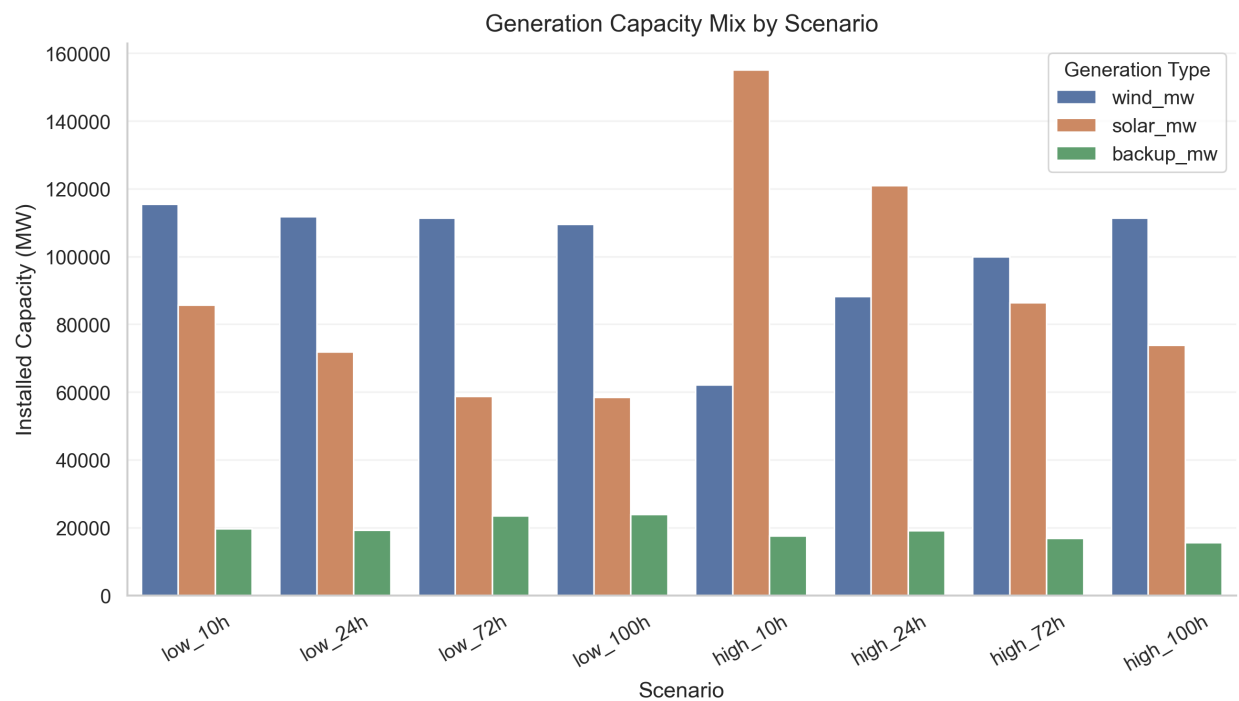


Figure 6: Installed capacity of generators by scenario.

Since backup generation is fully utilized under the emissions cap in the primary scenarios, the following sensitivity analysis examines how relaxing that constraint changes the balance among firm generation, LDES, and transmission.

6.7 CO₂ Emissions Constraint Sensitivity

Because the core scenarios are solved under a stringent emissions cap, a limited sensitivity analysis was performed to test whether the observed roles of transmission and LDES persist when additional dispatchable generation is allowed. In addition to the primary 5% baseline emissions cap, additional scenarios were solved with emissions caps equal to 10% and 20% of the gas-only baseline for both renewable correlation cases.

Relaxing the emissions constraint substantially reduced total system cost in both correlation cases. In the high-correlation system, system cost declined from approximately 120 \$/MWh at the 5% emissions cap to 92 \$/MWh at the 20% cap, representing a reduction of roughly 23%. In the low-correlation system, costs declined from approximately 107 \$/MWh to 91 \$/MWh over the same range, corresponding to a smaller reduction of roughly 15%. This indicates that highly correlated renewable systems benefit more from the availability of additional dispatchable generation.

The emissions sensitivity also produced substantial changes in storage deployment. As the emissions cap was relaxed, long-duration storage deployment declined significantly in both systems. In the high-correlation case, LDES power capacity declined from approximately 39 GW at the 5% cap to 22 GW at the 20% cap. In the low-correlation case, LDES deployment declined from approximately 27 GW to 13 GW over the same range. At the same time, backup generation capacity increased substantially as the system relied more heavily on dispatchable gas generation to meet reliability needs.

In contrast, transmission deployment remained comparatively persistent across emissions scenarios. In the low-correlation system, transmission capacity declined only modestly from approximately 32 GW to 29 GW as the emissions cap increased. In the high-correlation system, transmission deployment remained relatively stable and even increased between the 5% and 10% cases. This suggests that while the value of long-duration storage is highly sensitive to decarbonization constraints, transmission retains structural value across a wider range of system conditions.

Correlation case	Cost reduction	LDES change	Transmission change
Low	107 → 91 \$/MWh (-15%)	27 → 13 GW (-51%)	32 → 29 GW (-10%)
High	120 → 92 \$/MWh (-23%)	39 → 22 GW (-44%)	27 → 28 GW (+7%)

Table 7: Changes in major system outcomes between the 5% and 20% CO₂ cap scenarios.

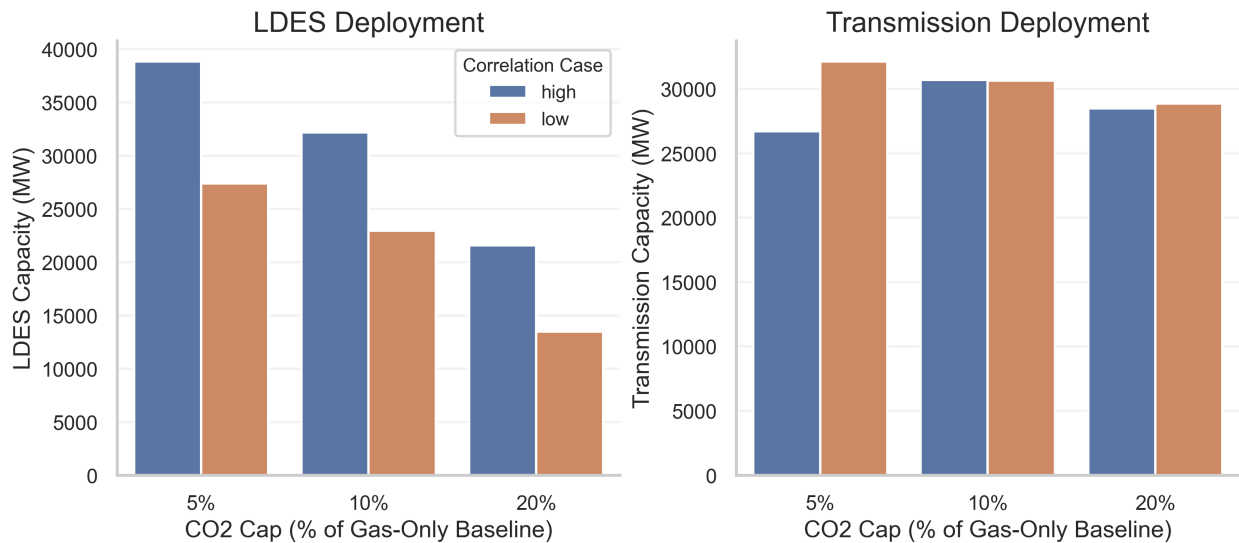


Figure 7: LDES and transmission capacities by CO₂ emissions constraint scenario.

Overall, the sensitivity analysis reinforces the distinct roles of the three major flexibility resources. When the emissions constraint is relaxed, the model substitutes additional gas generation primarily for LDES, while transmission remains comparatively stable. This suggests that LDES is more closely tied to the need for deep decarbonization balancing, while transmission continues to provide value by accessing spatial diversity.

7. Discussion

7.1 Core Findings

The results show consistent patterns across both renewable correlation cases and storage durations. Increasing LDES duration leads to:

- Lower system costs
- Reduced renewable curtailment
- Shifts in the balance of flexibility resources

These effects are present in both correlation cases, but are more pronounced under high renewable correlation, where temporal balancing plays a larger role.

The most important finding is not simply that longer-duration storage lowers cost and curtailment, but that it changes the relationship between storage and transmission differently depending on renewable correlation. In the low-correlation case, increasing LDES duration slightly reduces transmission deployment, suggesting partial substitution between temporal and spatial balancing. However, in the high-correlation case, transmission deployment rises as storage duration increases through the 72-hour case, suggesting complementarity between transmission and longer-duration storage.

A second key finding is that increasing LDES duration primarily displaces short-duration batteries, not firm backup capacity. Battery deployment falls sharply as LDES duration increases, especially in the high-correlation case, while backup capacity remains significant across all scenarios. This indicates that longer-duration storage changes the composition of storage resources but does not eliminate the need for firm capacity under the modeled emissions constraint.

The primary scenarios also show that the value of longer-duration storage is larger in high-correlation systems, but that spatial diversity remains important. Even when LDES duration increases, transmission remains a major component of the least-cost system, particularly in the low-correlation case. The CO₂ sensitivity analysis extends this finding by testing whether the same pattern holds when additional dispatchable generation is available. Relaxing the emissions constraint substantially reduces LDES deployment and system cost, especially in the high-

correlation case. Transmission deployment, however, remains comparatively persistent. This suggests that LDES is more sensitive to emissions constraints, while transmission provides broader structural benefits associated with spatial diversification.

7.2 Interpretation

These results support a distinction between spatial and temporal flexibility in high-renewable systems. Consistent with the central hypothesis, the system shifts toward spatial flexibility under low renewable correlation and toward temporal flexibility under higher correlation and longer storage durations. However, the results also show that the two strategies interact in more complicated ways than a simple substitution framing suggests. In the low-correlation case, transmission remains valuable because regions can share less-synchronized renewable output. In the high-correlation case, longer-duration storage becomes more important, but it also appears to increase the usefulness of transmission by allowing surplus generation to be moved and stored more effectively.

The results also highlight an important difference between energy adequacy and capacity adequacy. LDES reduces curtailment and improves system efficiency by shifting energy across time, but it does not eliminate the need for dispatchable backup capacity. Gas generation is dispatched up to the emissions cap in every scenario, meaning the model uses all allowable gas energy. However, backup capacity still varies across scenarios. This means gas is not primarily competing with storage on annual energy; instead, it provides dispatchable capacity during peak net-load periods. In this model, longer-duration storage improves energy balancing, but it does not fully solve the capacity adequacy problem.

This distinction between energy balancing and dispatchable capacity is further illuminated by the emissions sensitivity analysis. As emissions constraints were relaxed, the model reduced deployment of long-duration storage much more aggressively than transmission. This suggests that LDES is primarily addressing a deep-decarbonization balancing problem that becomes less critical when additional dispatchable generation is available. Transmission, by contrast, appears to retain value even under less strict emissions constraints because geographic diversity continues to reduce aggregate variability and improve renewable utilization.

7.3 Limitations

Several modeling assumptions influence the interpretation of these results and must be acknowledged. For one, LDES technologies of different durations are assigned identical costs per unit of power capacity. In reality, longer-duration storage systems typically exhibit higher costs per unit of energy and potentially different power-to-energy cost relationships. This assumption likely biases the model toward longer storage durations and may overstate the relative value of very long-duration storage. Likewise, not all characteristics of the modeled storage and generation technologies are fully captured. Importantly, ramping speeds of various technologies are not accounted for, so the utility of resources with fast ramping times (e.g. short-duration batteries) may be understated.

In addition, the model represents a stylized multi-region system rather than a fully detailed power system with a realistic transmission network, operational constraints, or reliability requirements. Transmission is represented as aggregated inter-regional capacity with a fixed capacity expansion cost and does not capture network congestion effects, distance losses, or non-technical constraints on expansion. Similarly, storage is modeled with simplified characteristics and does not include degradation, cycling limits, or other technology-specific constraints.

Finally, the temporal resolution and scope of the input data limit the model's ability to represent extreme events. Although hourly inputs capture broad variability, they do not reflect sub-hourly spikes or short-duration anomalies that can stress system operations. In addition, the use of a single historical year does not capture interannual variability and may omit anomalous years that drive capacity adequacy requirements.

7.4 Implications

Despite these limitations, the results provide several insights for both modeling and system planning. From a modeling perspective, the results highlight the importance of representing both transmission and storage when analyzing high-renewable systems. Treating these resources independently risks mischaracterizing how systems balance spatial and temporal variability.

From a planning perspective, the findings suggest that long-duration storage can play a significant role in reducing system costs and curtailment, particularly in systems with high renewable correlation. Furthermore, the results suggest that storage, transmission, and firm capacity serve different system functions: storage shifts energy across time, transmission accesses spatial diversity, and firm generation provides dispatchable capacity during peak net-load conditions. The emissions sensitivity analysis further suggests that transmission and LDES may respond differently to future policy environments, with the value of LDES appearing more sensitive to decarbonization stringency than the value of transmission.

8. Conclusions and Recommendations

8.1 Conclusions

This study examined the interaction between long-duration energy storage and transmission in high-renewable power systems using a PyPSA-based capacity expansion modeling framework. The results show that these resources provide distinct but interrelated forms of flexibility along temporal and spatial dimensions.

Increasing LDES duration reduces system costs and renewable curtailment while shifting the system toward greater reliance on temporal balancing. This shift primarily displaces short-duration storage and, to a lesser extent, reduces reliance on transmission. However, both transmission and dispatchable backup capacity remain present across all scenarios. While storage of increasing durations improves energy balancing over extended periods, it does not eliminate the need for firm capacity to meet peak demand.

Finally, the role of transmission is strongly influenced by renewable correlation, but not in a simple linear way. In the low-correlation case, transmission remains consistently high because spatial diversity provides a strong balancing benefit. In the high-correlation case, transmission deployment increases as LDES duration rises through the 72-hour case, suggesting that longer-duration storage can make transmission more useful rather than simply replacing it. This nuance is central to the study's main finding: spatial and temporal flexibility are not interchangeable resources, and their relationship depends on the structure of renewable variability.

The CO₂ sensitivity analysis reinforces this interpretation. Relaxing the emissions cap substantially reduced LDES deployment and system cost, particularly in the high-correlation system. Transmission deployment, however, remained comparatively stable across emissions scenarios, suggesting that spatial diversification retains value even when additional dispatchable generation is available. This result further supports the conclusion that transmission and storage serve structurally different roles within high-renewable systems.

8.2 Recommendations for Future Work

The robustness and applicability of this analysis could be significantly improved with additional work. Perhaps most importantly, future work should incorporate more realistic cost assumptions for long-duration storage, including explicit differentiation between power and energy costs and variation across storage durations. Similarly, future work should model constraints on storage behavior, including degradation and cycling limits, and more accurately represent the ramping behaviors of various storage technologies. Together, these improvements would provide a more accurate representation of the economic trade-offs between storage technologies.

Future work should also explore systems containing multiple classes of long-duration storage simultaneously rather than representing LDES as a single fixed-duration resource. Real-world LDES technologies span a wide range of discharge durations, efficiencies, ramping characteristics, and cost structures, and future systems will likely deploy portfolios of storage technologies optimized for different balancing needs. Modeling mixed-duration storage systems could improve understanding of how different storage technologies interact with one another, with transmission expansion, and with firm generation resources across multiple timescales.

The approach to transmission modeling could be significantly improved in extensions to this work. Future studies should explore how constraints on transmission expansion influence the relative value of long-duration storage. Geographic or otherwise physical constraints are likely relatively easy to model with the right tools, while other constraints, particularly non-technical and political, are likely much more difficult. More detailed transmission modeling that includes network constraints and greater spatial resolution would improve the representation of transmission value and limitations. This is particularly important for understanding congestion

and the geographic distribution of renewable resources. An argument could be made for modeling a real-world system; however, this approach risks coming to conclusions only relevant to the specific system modeled. A stylized model built with reasonable assumptions may continue to be the best approach.

Additionally, future models could incorporate more sources of flexibility, such as demand response, flexible loads, hydrogen production, or alternative firm low-carbon resources. This would allow for a more complete assessment of system flexibility options and may clarify which functions are most effectively served by transmission, storage, or firm generation. The CO₂ sensitivity analysis also suggests that future work should more explicitly compare LDES with other sources of firm or flexible capacity under varying decarbonization constraints.

Lastly, further work could explore sensitivity to temporal resource assumptions, including extreme weather events, interannual weather variability, and sub-hourly load events, to better understand system performance under stress conditions.

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10. Appendix

10.1 GitHub Repo

All code associated with this project can be found at the following GitHub repository:

<https://github.com/rcolcord/esm-capstone.git>

10.2 Model Results

10.2.1 Primary Scenario Analysis Results

scenario_id	corr high dur_10h	corr high dur_24h	corr high dur_72h	corr high dur_100h	corr low dur_10h	corr low dur_24h	corr low dur_72h	corr low dur_100h
renewable_correlation_case	high	high	high	high	low	low	low	low
ldes_duration_hours	10	24	72	100	10	24	72	100
ldes_capex_per_mw	1800000	1800000	1800000	1800000	1800000	1800000	1800000	1800000
transmission_capex_per_mw	400000	400000	400000	400000	400000	400000	400000	400000
co2_cap_tons	6576567.937	6576567.937	6576567.937	6576567.937	6576567.937	6576567.937	6576567.937	6576567.937
solver_status	ok	ok	ok	ok	ok	ok	ok	ok
solver_condition	optimal	optimal	optimal	optimal	optimal	optimal	optimal	optimal
objective	50977412996.894	45566517451.063	40281284363.309	39533104164.034	41868300757.301	38202615763.687	35576609067.394	35230777686.617
backup_mw	17625.901	19140.187	16916.609	15530.350	19745.849	19288.125	23527.665	23849.945
solar_mw	155164.768	120927.002	86338.045	73856.002	85758.897	71822.473	58706.786	58535.394
wind_mw	62201.665	88288.550	100008.385	111331.184	115416.282	111817.329	111405.617	109473.015
battery4h_mw	32876.039	0.000	0.000	639.240	0.000	0.000	562.573	401.571
ldes_mw	45773.621	50719.891	40996.012	38836.839	35833.750	31509.747	27447.497	27392.803

battery4h_mwh	131504.157	0.000	0.000	2556.960	0.000	0.000	2250.292	1606.284
ldes_mwh	457736.215	1217277.380	2951712.898	3883683.895	358337.497	756233.929	1976219.801	2739280.294
transmission_mw	20751.120	26582.515	32734.652	26709.977	36326.303	34435.257	31995.694	32119.423
vres_available_mwh	458609509.874	465378395.545	437055033.458	436097890.830	437492105.848	402638986.600	378653417.452	374483993.848
vres_dispatched_mwh	341415977.161	342729311.076	341389595.649	339808796.340	328240540.834	331158414.189	333833463.747	334585132.324
vres_curtailment_mwh	117193532.713	122649084.469	95665437.809	96289094.490	109251565.014	71480572.411	44819953.706	39898861.525
total_load_mwh	328828396.830	328828396.830	328828396.830	328828396.830	328828396.830	328828396.830	328828396.830	328828396.830
system cost per mwh	155.027	138.572	122.499	120.224	127.326	116.178	108.192	107.140
total generation mwh	357857397.003	359170730.918	357831015.490	356250216.181	344681960.676	347599834.030	350274883.588	351026552.165
generator_capex_total	34345032135.554	33372826747.101	29856007158.107	29670958296.632	32369475137.314	29666000363.638	27871977819.611	27560516553.845
backup_capex	1059816979.667	1150868547.670	1017168415.068	933814878.900	1187286030.197	1159763836.054	1414680550.622	1434058699.802
solar_capex	23471445114.641	18292370885.391	13060172834.806	11172040670.806	12972566363.571	10864432950.935	8880450969.307	8854524787.084
wind_capex	9813770041.246	13929587314.039	15778665908.232	17565102746.926	18209622743.546	17641803576.649	17576846299.681	17271933066.959
storage_capex_total	15103898545.146	10443594071.729	8441376859.260	8107205207.691	7378429433.300	6488085873.433	5748814033.686	5709741737.728
battery4h_capex	5678777274.667	0.000	0.000	110417866.623	0.000	0.000	97174909.525	69364567.036
ldes_capex	9425121270.479	10443594071.729	8441376859.260	7996787341.068	7378429433.300	6488085873.433	5651639124.161	5640377170.692
transmission_capex_total	788618423.328	1010232739.366	1244036453.077	1015076766.840	1380532293.822	1308665633.753	1215953321.231	1220655502.178
transmission_capex	788618423.328	1010232739.366	1244036453.077	1015076766.840	1380532293.822	1308665633.753	1215953321.231	1220655502.178
wind_opex	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

solar_opex	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
backup_opex	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867
generator_opex_total	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867
total_capex	50237549104.028	44826653558.195	39541420470.444	38793240271.163	41128436864.437	37462751870.824	34836745174.528	34490913793.752
total_opex	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867	739863892.867
reconstructed total cost	50977412996.895	45566517451.063	40281284363.311	39533104164.031	41868300757.304	38202615763.691	35576609067.395	35230777686.619
cost reconstruction error	-0.001	0.000	-0.002	0.004	-0.003	-0.004	-0.001	-0.002

10.2.2 CO₂ Emissions Sensitivity Analysis Results

scenario_id	corr_high_co2frac_0.05	corr_high_co2frac_0.1	corr_high_co2frac_0.2	corr_low_co2frac_0.05	corr_low_co2frac_0.1	corr_low_co2frac_0.2
renewable_correlation_case	high	high	high	low	low	low
co2_baseline_frac	0.05	0.1	0.2	0.05	0.1	0.2
ldes_duration_hours	100	100	100	100	100	100
ldes_capex_per_mw	1800000	1800000	1800000	1800000	1800000	1800000
transmission_capex_per_mw	400000	400000	400000	400000	400000	400000
co2_cap_tons	6576567.937	13153135.873	26306271.746	6576567.937	13153135.873	26306271.746
solver_status	ok	ok	ok	ok	ok	ok
solver_condition	optimal	optimal	optimal	optimal	optimal	optimal
objective	39533104164.034	35609765013.661	30280917431.559	35230777686.617	32857953334.507	29876974412.115
backup_mw	15530.350	24973.250	37513.924	23849.945	30085.405	40890.499
solar_mw	73856.002	60020.269	47672.303	58535.394	49988.921	40891.817
wind_mw	111331.184	99860.211	78125.273	109473.015	102130.621	91271.450
battery4h_mw	639.240	0.000	0.000	401.571	0.000	0.000
ldes_mw	38836.839	32185.806	21594.107	27392.803	22967.986	13483.601
battery4h_mwh	2556.960	0.000	0.000	1606.284	0.000	0.000
ldes_mwh	3883683.895	3218580.646	2159410.669	2739280.294	2296798.572	1348360.092
transmission_mw	26709.977	30700.148	28472.164	32119.423	30648.295	28856.165
vres_available_mwh	436097890.830	385429106.215	304508921.837	374483993.848	339509979.719	295902262.634
vres_dispatched_mwh	339808796.340	317811473.796	276424261.710	334585132.324	313616407.222	270459259.494
vres_curtailment_mwh	96289094.490	67617632.419	28084660.127	39898861.525	25893572.497	25443003.140

total_load_mwh	328828396.830	328828396.830	328828396.830	328828396.830	328828396.830	328828396.830
system_cost_per_mwh	120.224	108.293	92.087	107.140	99.924	90.859
total_generation_mwh	356250216.181	350694313.479	342189941.076	351026552.165	346499246.905	336224938.860
generator_capex_total	29670958296.632	26336027945.008	21793032062.393	27560516553.845	25484203160.040	23044507687.583
backup_capex	933814878.900	1501601241.238	2255651711.444	1434058699.802	1808986851.783	2458679707.138
solar_capex	11172040670.806	9079138711.509	7211288032.304	8854524787.084	7561718016.403	6185618302.880
wind_capex	17565102746.926	15755287992.260	12326092318.645	17271933066.959	16113498291.854	14400209677.565
storage_capex_total	8107205207.691	6627291423.167	4446383476.793	5709741737.728	4729275153.500	2776371405.292
battery4h_capex	110417866.623	0.000	0.000	69364567.036	0.000	0.000
ldes_capex	7996787341.068	6627291423.167	4446383476.793	5640377170.692	4729275153.500	2776371405.292
transmission_capex_total	1015076766.840	1166717859.748	1082046320.906	1220655502.178	1164747235.235	1096639747.771
transmission_capex	1015076766.840	1166717859.748	1082046320.906	1220655502.178	1164747235.235	1096639747.771
wind_opex	0.000	0.000	0.000	0.000	0.000	0.000
solar_opex	0.000	0.000	0.000	0.000	0.000	0.000
backup_opex	739863892.867	1479727785.734	2959455571.469	739863892.867	1479727785.734	2959455571.469
generator_opex_total	739863892.867	1479727785.734	2959455571.469	739863892.867	1479727785.734	2959455571.469
total_capex	38793240271.163	34130037227.923	27321461860.091	34490913793.752	31378225548.774	26917518840.646
total_opex	739863892.867	1479727785.734	2959455571.469	739863892.867	1479727785.734	2959455571.469
reconstructed_total_cost	39533104164.031	35609765013.657	30280917431.559	35230777686.619	32857953334.509	29876974412.115
cost_reconstruction_error	0.004	0.003	-0.001	-0.002	-0.001	0.000